PLC Commissioning & Troubleshooting Guide

Adaptive Demand Allocation Architecture (ADA)

Verification Protocols, Span-Injection Test Matrices, and Field Troubleshooting Checklists

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This document is the field commissioning and troubleshooting companion to *PLC Implementation in GX Works2*. Every register, flag, and threshold referenced below matches that document exactly – no new logic is introduced here. Where a test target depends on an item still flagged open in the parent documents, that dependency is called out explicitly rather than assumed resolved.

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1 I/O Domain Verification Protocols

I/O verification is the gate before any mathematical or arbitration logic can be trusted – every physical loop must be confirmed for electrical integrity, scaling accuracy, and actuation before higher-level testing begins.

1.1 Analog Inputs: Calibration and Scaling

All channels are 4–20mA loop-powered. Inject 4mA (0%), 12mA (50%), and 20mA (100%) with a certified calibrator and confirm scaled register values track linearly. Maximum allowable deviation: ±1.0% of full scale.

Table 1: Analog input verification – channel assignment and target registers.

Ch.	Correspondence	Target Register	4mA	12mA	20mA
AI-1	Level: T5	D_AlphaS (%), feeds $D_HeightT5 \rightarrow D_MassT5$	☐	☐	☐
AI-2	Level: TA	$D_AlphaTA$, feeds $D_HeightTA \rightarrow D_MassTA$	☐	☐	☐
AI-3	Level: TB	$D_AlphaTB$, feeds $D_HeightTB \rightarrow D_MassTB$	☐	☐	☐
AI-4	Level: TC	$D_AlphaTC$, feeds $D_HeightTC \rightarrow D_MassTC$	☐	☐	☐
AI-5	Temp: T5	D_ThetaS	☐	☐	☐
AI-6	Temp: TA	$D_ThetaTA$	☐	☐	☐
AI-7	Temp: TB	$D_ThetaTB$	☐	☐	☐
AI-8	Temp: TC	$D_ThetaTC$	☐	☐	☐
AI-9	Temp: Solar loop	$D_ThetaSol$	☐	☐	☐
AI-10	Temp: Thermal battery	D_ThetaH	☐	☐	☐
AI-11	Flow: Trunk/outlet	Flow rate register	☐	☐	☐

All four tanks (T5, TA, TB, TC) undergo height-to-mass conversion – this is not a T5-only calculation. Confirm the technician performing this test has the correct constants per tank (2.4353 kg/mm, T5; 4.0455 kg/mm, T10 tanks) before recording pass/fail.

1.2 Digital Inputs: Signal Loop Checks

For NC devices (E-STOP, float switches), confirm the bit reads HIGH (1) unactuated and drops LOW (0) when tripped.

Table 2: Digital input loop verification.

Tag	Correspondence	Type	Verified
X0	E-STOP	NC	☐ Pass ☐ Fail
X1	Automation Enable/Disable	NO	☐ Pass ☐ Fail
X2	Circulation Pump (manual)	NO	☐ Pass ☐ Fail
X3	Oil Distribution Pump (manual)	NO	☐ Pass ☐ Fail
X4	Jacket Pump (manual)	NO	☐ N/A – intentionally unused, see §1.4
X5	AODD Pump (manual)	NO	☐ Pass ☐ Fail
X6	Float – HEA	NC	☐ Pass ☐ Fail
X7	Float – SOL	NC	☐ Pass ☐ Fail
X10	Loop Exit	NO	☐ Pass ☐ Fail
X_Confirm	PO designation confirm	NO	☐ Pass ☐ Fail (tag TBD)
X_EStopReset	E-STOP reset	NO	☐ Pass ☐ Fail (tag TBD)
X_AlarmReset	Master alarm acknowledge	NO	☐ Pass ☐ Fail (tag TBD)

1.3 Digital Outputs: Actuation Checks

Force each coil via the GX Works2 device test menu; confirm the field component actuates and the correct valve state results (recall: all solenoid/pneumatic valves are **normally open**, energise-to-close).

Table 3: Digital output actuation verification.

Tag	Correspondence	Field Component	Verified
Y0	E-STOPPED lamp	Panel indicator	☐
Y1	Automation Enabled lamp	Panel indicator	☐
Y2	Circulation Pump – HSM	VFD-01, 2-destination speed	☐
Y3	Oil Distribution Pump active	VFD-02	☐
Y4	Jacket Pump active	Continuous, gated on M2_SystemOK only	☐
Y5	AODD Pump active	Air solenoid, $T_{10} \rightarrow T_5$ transfer	☐
Y6	Circulation Pump – LSM	VFD-01, 1-destination speed	☐
Y7	Heating Element 1	12kW bank 1 contactor	☐
Y10	Heating Element 2	12kW bank 2 contactor	☐
Y11–Y14	SV100–SV103	T5/TA/TB/TC branch valves	☐
Y15/Y16	SV104/SV105	Solar selector: through/bypass	☐
Y17	Circulation Pump – FSM	VFD-01, 3-destination speed	☐
Y20	Heating Element 3	12kW bank 3 contactor	☐
Y21–Y24	PBV100–PBV103	T5/TA/TB/TC oil valves	☐
Y25	Solar Cycle Active lamp	Panel indicator, mirrors Y15	☐

1.4 Auto/Manual Mode Gating Verification

Jacket Pump (Y4) is a deliberate exception. It runs continuously off `M2_SystemOK` alone – no manual/auto split, no dependency on X4. X4 is intentionally unmapped; confirm it drives nothing during commissioning, not that it "should."

Table 4: Mode selection state verification.

Test Condition	X1	M2_SystemOK	M3_AutoEnabled	Y1
Manual selection	0	1	0	0
Auto selection	1	1	1	1
Interlock drop (E-STOP active)	1	0	1	0
Safety override	0	0	0	0

Table 5: Manual-mode input-to-output actuation. Set X1=0, M3_AutoEnabled=0, M2_SystemOK=1.

Device	Tag	Expected Behaviour	Verified
Circulation Pump switch	X2	Y2 (HSM) energises on HIGH	☐
Oil Distribution switch	X3	Y3 energises on HIGH	☐
Jacket Pump override	X4	**No change** – Y4 always on via M2	☐
AODD Pump switch	X5	Y5 energises on HIGH	☐

Auto-mode isolation cross-check: flip to Auto (X1=1). Confirm X2, X3, X5 toggling has **zero** effect on Y2, Y3, Y5 while automation governs.

2 Mathematical Domain Verification Protocols

2.1 Span Injection Methodology

Every mathematical block undergoes a **4-point linear span injection test**: simulated inputs at 25%, 50%, 75%, 100% of the designed range, verifying fixed-point scaling holds linearly and that 32-bit (`DMUL/DADD/DDIV`) blocks handle full-scale values without truncation or overflow.

2.2 Acceptance Criteria

Maximum allowable deviation between PLC register value and theoretical value: ±5%.

$$\Delta = \left| \frac{\text{PLC Value} - \text{Theoretical Value}}{\text{Theoretical Value}} \right| \times 100\% \tag{1}$$

2.3 10-Second Refresh Oscillator

The planning window is a fixed constant (5 minutes, K300 seconds), not HMI-adjustable. This was proposed during development, not user-confirmed, and there is currently no mechanism to change it from the HMI – do not test for HMI-driven window adjustment; that feature does not exist in the current design.

☐ Confirm T0 self-resets every 10.0s continuously while M2_SystemOK=1.

☐ Confirm M10_Refresh pulses for exactly one scan per T0 cycle.

2.4 Mode Register (Eco/Efficiency/Standard/Full Throttle)

Driven by T5's own level (*D_AlphaS*), **not** a system-wide state – $D_Mode \in \{0, 1, 2, 3\}$ governs P3's operating posture only. Four-tier hysteresis ladder, 5% deadband at each transition.

Table 6: Mode register hysteresis verification.

Mode	D_Mode	Rising/Falling Threshold	Verified
Eco	0	Enter $\geq 90\%$, exit $< 85\%$	☐
Efficiency	1	Enter $\geq 70\%$, exit $< 65\%$	☐
Standard	2	Enter $\geq 50\%$, exit $< 45\%$	☐
Full Throttle	3	Settled $< 45\%$	☐

☐ Step *D_AlphaS* slowly across 90%, 70%, 50% and their exit points; confirm no chatter within any deadband.

☐ Confirm power-up default is deliberately chosen (open item – see §4) and does not simply default to whatever the register happens to power up as.

2.5 Suspicion Score (PO Tank Overrun)

This is event-driven, not a continuous rate-of-change counter. *D_Suspicion* increments by exactly 1 each time a PO dispense crosses its own $2 \times n_d$ ceiling – not once per anomalous scan cycle. It caps at 3, resets every 24 hours, and auto-increments once on a 5-minute unanswered HMI prompt.

Table 7: Suspicion meter tier verification.

D_Suspicion	Expected Action	Verified
1	M_HMIPrompt fires	☐
2	M_Alarm fires	☐
3	M_DisableFeed **and** M_Alarm fire	☐

☐ Force one overrun event; confirm *D_Suspicion* increments by exactly 1, not more.

☐ Leave an M_HMIPrompt unanswered 5 minutes; confirm auto-increment fires and M_PromptActive clears.

☐ Confirm reset to 0 occurs on the 24hr rollover, independent of accumulated count.

2.6 E_{total} Calculation

Sum of remaining energy demand across every **currently-served** branch: E_α (per active T10 melt tank) $+E_5 + E_{C^*}$. Loss terms are power ($UA\Delta T$, Watts) multiplied by the 5-minute planning window to convert to Joules before summing – confirm this conversion is present, not silently dropped.

Table 8: E_{total} component verification.

Component	Gating Condition	Verified
E_α (TA/TB/TC)	M_Serve_TA/TB/TC=1 AND not the PO tank	☐
E_5	Always summed if T5 branch is live	☐
E_{C^*}	Only if a PO tank is designated (M1/M2/M3)	☐

☐ Confirm *D_Etotal* only includes branches with live `M_Serve_*` flags – an idle T10 tank contributes 0, not a stale prior value.

☐ At 100% span (largest mass, largest $\Delta\theta$), confirm the intermediate 32-bit products do not overflow (theoretical ceiling worked in the parent document: $\approx 1.4 \times 10^9$ against a $\approx 2.15 \times 10^9$ 32-bit limit – narrow but sufficient at design masses).

☐ Confirm *D_EC_star* correctly zeroes when no PO tank is designated (M0=1).

2.7 T_f Thermal Blending

T5's own safety cutoff – a mass-weighted blend between the thermal buffer (θ_H) and T5's own state (θ_S), **not** downstream infrastructure protection.

☐ Confirm *D_Mp* is computed as T5's own remaining capacity to reach full (4871 kg − *D_MassT5*), never read from a T10 tank.

☐ Confirm $M_TfInterlock$ fires at exactly $D_Tf \geq 55.0\,°C$ (register value ≥ 550).

☐ Confirm the block runs every scan (unconditional), not gated on M10_Refresh – this is a deliberate deviation from the 10s pattern given T_f is a safety interlock.

2.8 Dynamic Power Evaluation

Two independent mechanisms, easy to conflate. D_Pbase (from D_Etotal/window) reflects actual demand. The three-condition taper against D_Twater is a **separate, protective ceiling check** on *measured* water temperature, independent of demand – do not test it as if it were tracking a target.

Table 9: Dynamic power taper verification.

Condition	Expected D_Ereq_kW	Verified
$D_Twater < 40.0\,°C$	Full D_Pbase	☐
$D_Twater \geq 70.0\,°C$	0 (safety taper)	☐
$40.0 \leq D_Twater < 70.0\,°C$	Linear interpolation	☐

☐ Confirm the fast override ($D_ThetaS < D_Twater_target$ forcing $D_Ereq_kW = 0$) is T5-only in the current build – do not expect it to fire for PO tank or T10 conditions; this extension is an open item (see §4).

2.9 Auxiliary Stage Register

Table 10: Auxiliary staging verification, gated on M2_SystemOK.

$D_AuxStage$	Threshold (rising/falling)	Verified
0→1	$>$ 0kW / $\leq$ 2kW	☐
1→2	$\geq$ 14kW / $<$ 10kW	☐
2→3	$\geq$ 26kW / $<$ 22kW	☐

☐ Confirm Y7/Y10/Y20 all drop instantly when M2_SystemOK=0 (E-STOP), regardless of $D_AuxStage$ value.

2.10 Tank Scoring and PO Exclusion Ranking

$S_{\text{casual}} = 0.6\alpha + 0.4\theta_{\text{norm}}$, $S_{\text{immediate}} = 0.4\alpha + 0.6\theta_{\text{norm}}$. The currently-designated PO tank is excluded via a -1 sentinel before ranking.

☐ Confirm the PO-designated tank never wins the ranking regardless of its own score.

☐ Confirm $D_RankedTankTemp$ updates only from the winning tank's own θ, never a stale value from a previous designation.

☐ Confirm the max-finding "challenger" pattern does not exhibit the compare-after-overwrite bug (i.e. each challenge uses the pre-update max, not a value already overwritten in the same rung group).

2.11 Need-Array Arbiter

Single-pass resolution, not a two-stage kernel. There is no separate "priority selection kernel" feeding into the arbiter – demand flags are computed and contention resolved in one block.

Table 11: Need-array combinatorial verification (7 profiles).

Simulated Profile	Solved Count	Trunk State	Verified
T5 only	1	LSM (Y6)	☐
PO only	1	LSM (Y6)	☐
One T10 melt tank only	1	LSM (Y6)	☐
Any 2 of {T5, PO, TA, TB, TC}	2	HSM (Y2)	☐
Any 3	3	FSM (Y17)	☐
All 4 destinations, PO designated	3 (tie-break)	FSM, lower $S_{\text{immediate}}$ of the two non-PO/non-T5 tanks dropped	☐
All 4 destinations, no PO tank	3 (tie-break)	FSM, lowest $S_{\text{immediate}}$ of the three T10 candidates dropped	☐

☐ Force the 4-way tie with two equal $S_{\text{immediate}}$ scores; confirm a deterministic (if arbitrary) outcome rather than an oscillating one – equal-score tie-break behaviour is an open item (§4), but it must not chatter between scans.

☐ Confirm the manual path (X2 only, HSM) has no LSM/FSM equivalent – this is a known limitation, not a defect (see §4).

2.12 Solar and Auxiliary Gating

Table 12: Solar selector verification.

Mode	Condition	Verified
Auto, through-solar	$D_ThetaSol > D_Twater_target \rightarrow$ Y15	☐
Auto, bypass	$D_ThetaSol \leq D_Twater_target \rightarrow$ Y16	☐
Manual, through-solar	X1=0, X10=1 → Y15	☐
Manual, bypass	X1=0, X10=0 → Y16	☐

☐ Confirm exactly one of Y15/Y16 is ever active, never both, never neither.

☐ Confirm Y25 mirrors Y15's actual state under both manual and automatic paths.

2.13 Oil Distribution

Priority cascade X>Y>Z, per the confirmed flow-logic table.

Table 13: Oil distribution verification.

Port	Valves Closed	Verified
X	PBV101 (Y22)	☐
Y	PBV100 (Y21) + PBV103 (Y24)	☐
Z	PBV100 (Y21) + PBV102 (Y23)	☐

Open item: X does not open a T5 outlet valve (Y21) while Y and Z both do. Confirm during commissioning whether this asymmetry is intentional (X routes from a source that bypasses T5 entirely) or a missing rung – do not assume either without checking against the physical P&ID.

2.14 Fault Detection and Master Alarm

Two independent mechanisms per tank, plus one hard limit:

Table 14: Fault detection verification, per tank (TA/TB/TC).

Check	Basis	Verified
Heating rate anomaly	Observed $\Delta\theta$ vs. live-mass-based max coil rate	☐
Cooling rate anomaly	Observed $\Delta\theta$ vs. live-mass-based max loss rate	☐
Level/mass anomaly	Any mass change while no active transfer (AODD or oil-distribution flag) is live	☐
High-limit hard alarm	$\theta > 75.0\,^{\circ}\mathrm{C} \rightarrow$ alarm + HE1/HE2/HE3 cutoff	☐

Open item surfaced during ladder review: `M_ServeTA_AODD` **(and TB/TC equivalents) are consumed by the level-anomaly check but are not produced by any block in the current program.** Until this is resolved, expect the level-anomaly check to behave as if no AODD transfer is ever active – confirm this explicitly rather than assuming it self-resolves.

☐ Confirm alarm latches (SET, not a momentary OUT) and requires X_AlarmReset **and** all three high-limit flags clear before it releases.

☐ Confirm E-STOP bypasses the 10s heartbeat entirely – trip within one scan, not waiting for M10_Refresh.

3 Known Open Items – Pre-Commissioning Sign-Off Checklist

Every item below is an unresolved design decision or gap carried forward from the architecture and GX Works2 reference documents, not a new finding. None should block commissioning of the items already confirmed above, but each should be explicitly signed off – resolved, accepted as a known limitation, or scheduled – before final hand-over.

Table 15: Consolidated open items.

Item	Sign-off
E-STOP reset input (X_EStopReset) has no assigned physical tag	☐
Mode register power-up default value not specified	☐
$M_ServeTA/TB/TC_AODD$ consumed but never produced	☐
Fast override (Dynamic Power) implemented for T5 only, not PO/T10	☐
Planning window (5 min) is a proposed value, not user-confirmed	☐
Tie-break on equal $S_{\text{immediate}}$ scores has no defined rule	☐
Manual trunk control has no LSM/FSM selection, HSM only	☐
Oil distribution: X lacks a T5 outlet valve rung; Y/Z both have one	☐
D128+ register range flagged as EEPROM-maintenance – write-endurance under continuous 10s-cycle use unconfirmed	☐
RTC/day-rollover source for the Suspicion Meter's 24hr reset unconfirmed	☐
ST language availability on the Coolmay L01S unconfirmed	☐

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